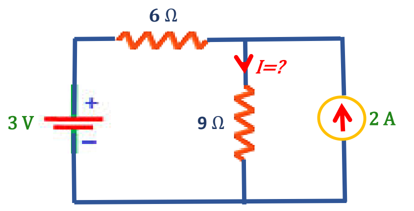
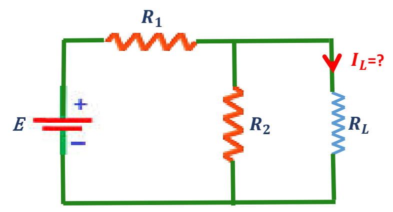
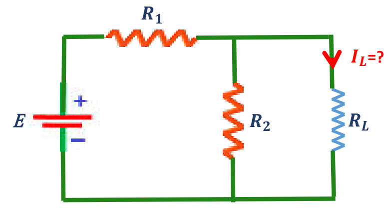
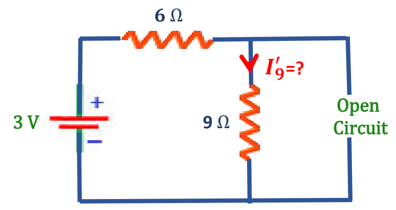
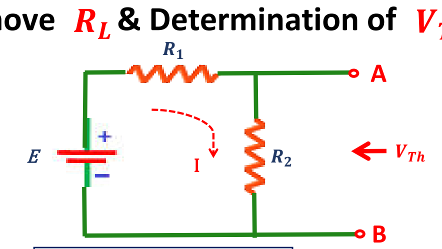
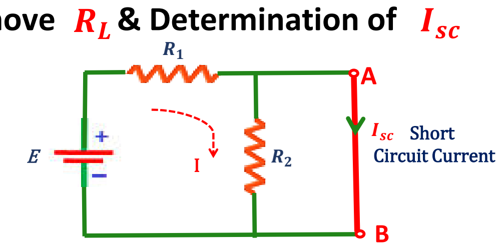
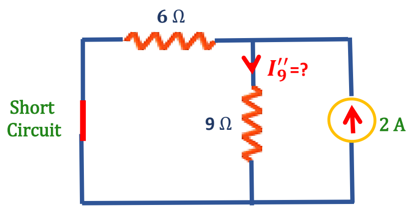
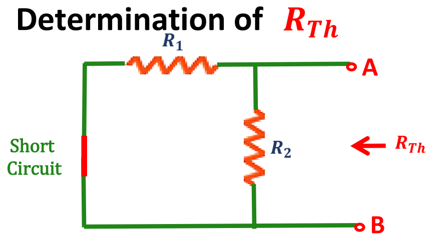
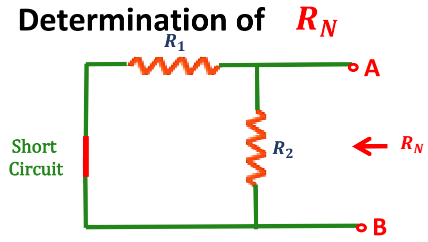
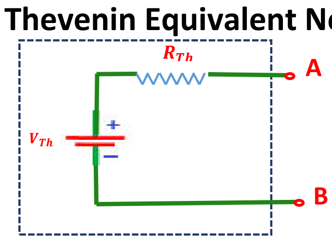
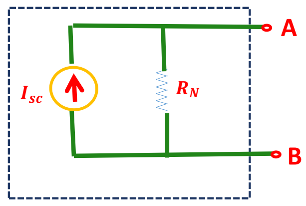
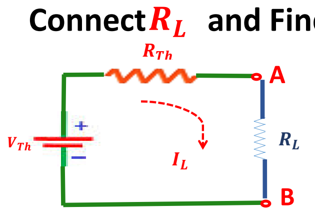
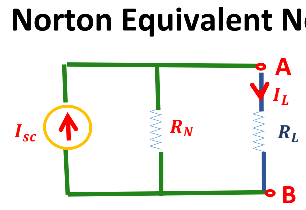
D.C. Circuits Theorems

Parameter	Superposition Theorem	Thevenin Theorem	Norton Theorem
Example			
Step 1	Consider 3 V alone  $I_9' = \frac{3}{6+9} = 0.2\ A$	Remove R_L & Determination of V_{Th}  $V_{Th} = I \times R_2$ $V_{Th} = \frac{E}{R_1 + R_2} \times R_2$	Remove R_L & Determination of I_{sc}  $I_{sc} = \frac{E}{R_1}$ Here, R_2 is omitted parallel with short circuit
Step 2	Consider 2 A alone  $I_9' = 2 \times \frac{6}{6+9} = 0.8\ A$	Determination of R_{Th}  $R_{Th} = R_1\ II\ R_2$ $R_{Th} = \frac{R_1 \times R_2}{R_1 + R_2}$	Determination of R_N  $R_N = R_1\ II\ R_2$ $R_N = \frac{R_1 \times R_2}{R_1 + R_2}$
Step 3	Consider 2 A alone $I_9 = I_9' + I_9''$ $= 0.2\ A + 0.8\ A$ $\boxed{I_9 = 1\ A}$	Thevenin Equivalent Network  V_{Th} = Thevenin Equivalent Voltage source R_{Th} = *Thevenin Equivalent Resistance*	Norton Equivalent Network  I_{sc} = Norton Equivalent Current source R_N = *Norton Equivalent Resistance*
Step 4	×	Connect R_L and Find I_L  $I_L = \frac{V_{Th}}{R_{Th} + R_L}$	Norton Equivalent Network  $I_L = I_{sc} \times \frac{R_N}{R_N + R_L}$
Statement	" In a linear electrical network having more than one independent energy source, the responce in any branch is equal to the algebraic sum of all the responses due to individual source acting alone with all other sources reduced to zero. "	" Any network having number of energy sources and resistances when viewed from open its output terminal A and B and can be replaced by simple equivalent network consisting of V_{Th} in seies with R_{Th}. "	" Any network having number of energy sources and resistances when viewed from open its output terminal A and B and can be replaced by simple equivalent network consisting of I_{sc} in parallel with R_N. "
Application	To determine voltage across any branch or current through in any branch of given network	To determine current in any branch of given network	To determine current in any branch of given network

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